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Appendicitis

Costi D. Sifri
Lawrence C. Madoff

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SHORT VIEW SUMMARY

Definition

- Appendicitis is acute inflammation of the vermiform appendix that is often related to obstruction and may be complicated by polymicrobial infection.
- Complications include perforation, peritonitis, and intraabdominal abscesses.

Epidemiology

- Acute appendicitis is a relatively common disease, often presenting in adolescence and early adulthood.
- Lifetime risk for appendicitis is 8.6% in men and 6.7% in women in industrialized countries.
- 190,000 appendectomies are performed annually in the United States.

Biology

- The microbial community of the appendix resides in a mucus-rich biofilm that is continuously shed into the intestinal tract and may serve to repopulate the gut after acute diarrheal illness.

Microbiology

- The infection is usually polymicrobial, frequently involving aerobic and anaerobic gram-negative bacilli.
- The human appendiceal microbiome is a highly diverse community of organisms, with notable differences in taxonomic composition compared with other locations in the lower intestinal tract.
- Genomic analysis has shown that several taxa, including Fusobacteriota, are enriched in the inflamed appendix.
- Infections by *Yersinia* spp., *Campylobacter* spp., parasites including *Entamoeba histolytica*, and viruses may mimic appendicitis (e.g., mesenteric adenitis) or cause obstruction, leading to acute appendicitis.

Diagnosis

- Clinical diagnosis is enhanced using imaging, especially computed tomography and ultrasonography.
- Diagnosis is more difficult in women of childbearing age, in whom gynecologic processes may mimic appendicitis.

Therapy

- Surgical removal of the appendix, often performed laparoscopically, is curative.
- Adjunctive use of broad-spectrum antibiotics, such as piperacillin-tazobactam or ceftriaxone and metronidazole, may be required.

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In 1886 Fitz first described the natural course of acute inflammation of the appendix. In his seminal report he coined the term *appendicitis* and advocated early surgical intervention. [1] With an incidence of 108 cases per 100,000 person-years in the United States and 100 to 150 cases per 100,000 person-years in most upper middle-income countries, appendicitis is

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Epidemiology

The lifetime risk for acute appendicitis is 8.6% in men and 6.7% in women in industrialized countries. [3] In 2018, a total of 190,000 appendectomies were performed in the United States, constituting 1.3% of all operating room procedures; [6] however, this represents a 42% decrease since 2011, likely because of the increasing adoption of nonoperative medical management of uncomplicated appendicitis. [7] Although the frequency of appendectomy surgery is declining, it is still ranked as the most common surgery for individuals aged 17 years and younger in the United States, the second most

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Similar to diverticulitis (see **Chapter 81**), the incidence of appendicitis is significantly lower in rural, nonindustrialized areas. For instance, individuals living in rural areas of sub-Saharan Africa have an estimated lifetime risk of appendicitis of less than 1%. [9, 10] Also like diverticulitis, appendicitis rates increase as societies become industrialized. [11–13]

Pathogenesis

The appendix is a tube-shaped structure, usually 5 to 10 cm long in adults, which arises 2 to 3 cm below the terminal part of the ileum along the medial posterior wall of the cecum. [14] Previously considered a vestigial organ, recent studies suggest that the appendix may contribute to the development and homeostasis of the gut microbiome and mucosal immune system. [15] Bacteria reside in the appendix within a mucus-rich biofilm that is intermittently shed into the intestinal tract at a rate of 2 to 3 mL per day. [14, 15] Given this, the appendix has been hypothesized to act as a microbial “sanctuary” that can act to repopulate the intestinal tract when needed, for example, during recovery from acute diarrheal illnesses. [15–17] Microbe-host interactions between the resident microbiota and gut-associated lymphoid tissue of the appendix are also hypothesized to aid in the maturation of normal mucosal immune responses. [15, 17, 18]

Typically, the appendix lies in an ascending retrocecal position, but atypical positions, such as descending pelvic, transverse retrocecal, and ascending postileal, are common and can alter the typical clinical features of acute appendicitis. [14] Physical obstruction of the appendiceal lumen by fecaliths or other causes (e.g., foreign bodies, tumor, stricture, or parasites) has classically been thought to be the primary pathogenic mechanism of acute appendicitis. [19–23] Mucus accumulates within the obstructed appendiceal lumen, and intraluminal pressure increases, leading to compression of lymphatic and vascular drainage and causing ischemic damage of the mucosa, followed by microbial invasion and inflammation. Continued inflammation and ischemia, if left untreated, may lead to gangrene and eventual perforation. The pathologic hallmark of acute appendicitis is the presence of polymorphonuclear cells within the appendiceal wall, accompanied by other markers of inflammation, such as edema and vascular congestion.

In recent years, this classic model of pathogenesis of acute appendicitis has faced challenges. Most modern case series report that fecaliths or other causes of obstruction are found in only a minority of cases. [24, 25] Page 997 The efficacy of medical rather than surgical treatment of acute appendicitis (see later) also suggests that physical obstruction may not be the primary pathogenic mechanism. This has led to the speculation that other inciting etiologies or cofactors, such as primary infectious agents, socioeconomic factors, genetic susceptibility, environmental factors, dietary fiber, or microbiome

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Among these factors, dietary considerations have received the longest scrutiny. Interest into the role of dietary fiber in influencing appendiceal inflammation dates back over a century. Short, noting high rates of appendicitis in Great Britain compared with Africa in the early 20th century, as well as Great Britain at the end of the 19th century, suggested that "the cause is the relatively less quantity of cellulose [fiber] eaten." [33] Fifty years later, Burkitt and colleagues hypothesized that dietary fiber acted as a bulking agent, increasing bowel transit time, preventing formation or impaction of fecaliths, and otherwise addressing mechanical issues of bowel function that reduce risk for appendicitis. [26, 34] However, they also noted that the quantity and variety of fecal bacteria can be influenced by the amount of dietary fiber consumed and that these changes might have implications for the development of appendicitis and other health issues common in Western societies. [34] In recent years, the influence of diet, including dietary fiber, and the gut microbiome has been increasingly recognized. [35, 36] This recognition could provide a plausible link between dietary fiber and modern understandings of the pathogenesis of appendicitis, as is discussed later.

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Recently, culture-independent 16S ribosomal RNA gene amplicon sequencing-based studies have demonstrated that the microbial community of the appendix is remarkably diverse, consisting of more than a dozen phyla and hundreds of different species of bacteria. [42–45] Furthermore, these studies have shown that the appendiceal microbiome is distinct, with both quantitative and qualitative differences in composition compared with elsewhere in the digestive tract. [42–44] Although the diversity of bacteria in the healthy appendix is large, a relatively small number of taxa predominate, including those of the phyla Firmicutes, Proteobacteria, and Bacteroidetes. [42–45] A few taxa, including *Fusobacterium*, *Peptostreptococcus*, and *Parvimonas*—all typically considered to be oral commensal bacteria and occasional periodontal pathogens—are enriched in the inflamed appendix and are associated with disease severity. [43–46] This has led some investigators to hypothesize that rather than being caused by physical obstruction of the appendiceal lumen, acute appendicitis may be the consequence of dysbiosis (microbial imbalance) of the appendix. [42, 43]

In some instances, intestinal pathogens have been associated with acute appendicitis. For example, *Yersinia enterocolitica* and *Yersinia pseudotuberculosis* are believed to have a causative role in some cases of acute appendicitis. [47–49] More

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Clinical Manifestations

The clinical manifestations of acute appendicitis are distinctive and, in many cases, diagnostic. Appendicitis classically starts

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Conventional cultures of inflamed or gangrenous appendixes typically yield 10 to 14 different organisms, which generally reflect the colonic microbiota. A mixture of colonic anaerobic and facultative bacteria is usually recovered, predominantly *Escherichia coli*, *Bacteroides fragilis* group, pigmented *Prevotella* spp., *Bilophila wadsworthia*, *Peptostreptococcus* spp., Enterobacteriaceae, and viridans streptococci, particularly the *Streptococcus anginosus* group. [37–39] *Pseudomonas aeruginosa* has also been found in a minority of cases (4%–15%) but is less frequently sensitive to routinely used antibiotics. [40, 41]

Recently, culture-independent 16S ribosomal RNA gene amplicon sequencing-based studies have demonstrated that the microbial community of the appendix is remarkably diverse, consisting of more than a dozen phyla and hundreds of different species of bacteria. [42–45] Furthermore, these studies have shown that the appendiceal microbiome is distinct, with both quantitative and qualitative differences in composition compared with elsewhere in the digestive tract. [42–44] Although the diversity of bacteria in the healthy appendix is large, a relatively small number of taxa predominate, including those of the phyla Firmicutes, Proteobacteria, and Bacteroidetes. [42–45] A few taxa, including *Fusobacterium*, *Peptostreptococcus*, and *Parvimonas*—all typically considered to be oral commensal bacteria and occasional periodontal pathogens—are enriched in the inflamed appendix and are associated with disease severity. [43–46] This has led some investigators to hypothesize that rather than being caused by physical obstruction of the appendiceal lumen, acute appendicitis may be the consequence of dysbiosis (microbial imbalance) of the appendix. [42, 43]

In some instances, intestinal pathogens have been associated with acute appendicitis. For example, *Yersinia enterocolitica* and *Yersinia pseudotuberculosis* are believed to have a causative role in some cases of acute appendicitis. [47–49] More commonly, however, nonplague *Yersinia* causes ileocecitis or mesenteric adenitis, which mimics acute appendicitis with fever, leukocytosis, and acute right lower quadrant pain. [50–52] Similarly, *Campylobacter* and nontyphoidal *Salmonella* can also cause pseudoappendicitis with ileocecitis and mesenteric adenitis. [51, 53] Appendiceal or ileocecal tuberculosis, actinomycosis, and histoplasmosis are more likely to cause subacute or recurrent disease rather than classic acute appendicitis. Viral causes of mesenteric adenitis and, rarely, appendicitis include measles, Epstein-Barr virus, cytomegalovirus, and adenovirus. [54–57] Amebiasis may also cause appendicitis. [58] In many cases of mesenteric adenitis, an infectious cause is not identified but is probably present. [59] The disorder is usually discovered at the time of surgery for suspected appendicitis; mesenteric lymph nodes of the right iliac fossa are enlarged, the adjacent bowel is only mildly inflamed, and the appendix appears normal.

Clinical Manifestations

The clinical manifestations of acute appendicitis are distinctive and, in many cases, diagnostic. Appendicitis classically starts as colicky, visceral periumbilical pain that evolves for the next 6 to 24 hours to localized, somatic right lower quadrant abdominal pain after inflammation extends to the parietal peritoneum. If the inflamed appendix lies in the anterior position, tenderness is often maximal at or near the McBurney point, which lies two to three fingerbreadths above the right anterior superior iliac spine on a line with the umbilicus. If the appendix lies in a position relatively hidden from the parietal peritoneum, pain may remain poorly localized, and migration to the right lower quadrant may be delayed or absent. [60] Pelvic appendixes can cause pelvic or left lower quadrant pain. Third-trimester pregnancy or intestinal malrotation may shift pain to the right upper quadrant. Pain is often accompanied by mild fever, anorexia, nausea, and vomiting.

Guarding is usually seen on examination of the abdomen. Rebound tenderness in the right lower quadrant with palpation of the left lower quadrant, known as the Rovsing sign, may be elicited. Other maneuvers that support the diagnosis of appendicitis include pain with active extension of the right hip, termed the *psoas sign*, and pain with internal rotation of the right hip, termed the *obturator sign*. High fever or a sudden reduction in pain suggests perforation, whereas abdominal rigidity suggests diffuse peritonitis. A palpable right lower quadrant mass may indicate a phlegmon or walled-off

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